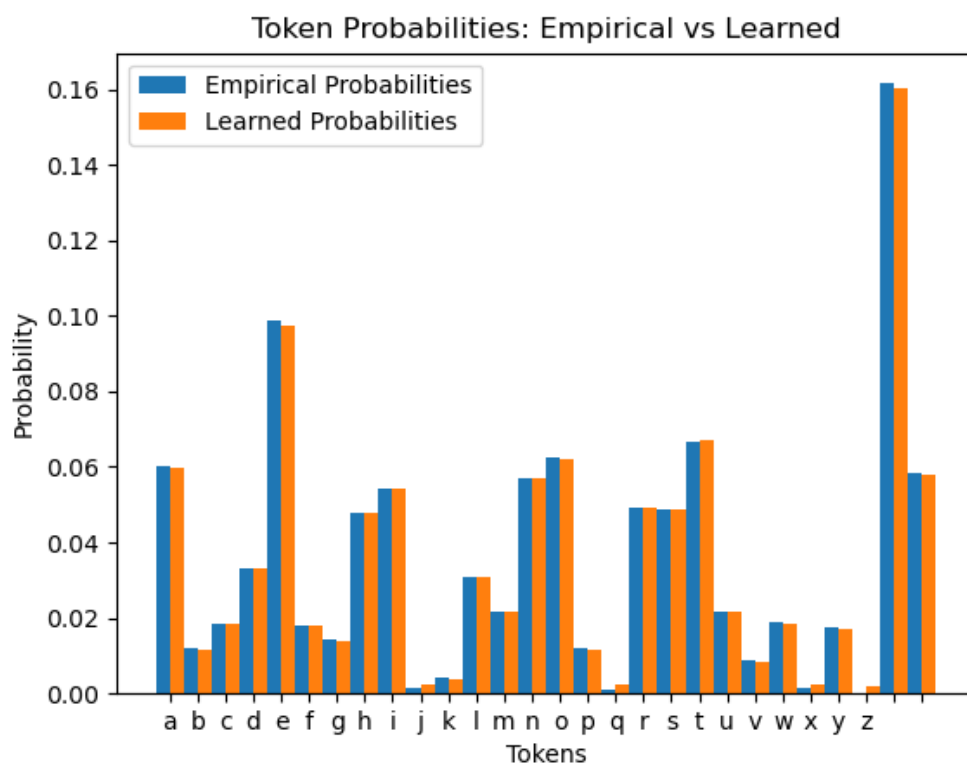
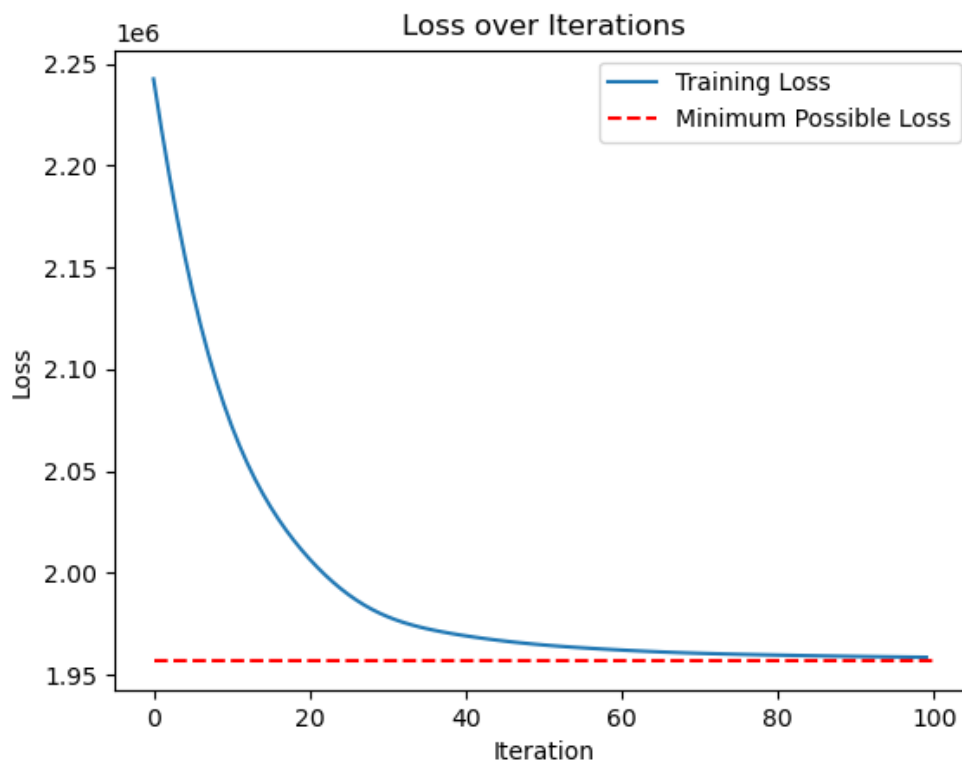


Explanation of the Neural Network:

The neural network in the provided code is designed to learn the unigram probability distribution of tokens (in this case, characters) from a text corpus. It aims to estimate the probability of each token appearing in the text by maximizing the likelihood of the observed data.

- **Inputs:** The inputs to the learned function are one-hot encoded vectors representing the tokens in the vocabulary. Each vector indicates the presence of a specific token at a particular position in the text.
- **Outputs:** The output of the learned function is the log-probability of the data given the model's estimated token probabilities. Specifically, it outputs the total log-likelihood of the observed data under the current model parameters.





Modifying the Code for Document Classification:

To adapt this code for document classification, we need to:

1. **Change the Model Architecture:**
 - Instead of learning unigram probabilities, the model should learn features that are predictive of document classes.
 - Replace the unigram model with a neural network suitable for classification, such as a simple feedforward network or an LSTM for sequence data.
2. **Prepare Labeled Data:**
 - Obtain a dataset where each document is associated with a class label.
 - Tokenize each document and represent it using suitable features (e.g., word embeddings).
3. **Modify the Loss Function:**
 - Use a classification loss function like cross-entropy loss.
 - The loss should measure the difference between the predicted class probabilities and the true class labels.
4. **Adjust the Training Loop:**
 - For each document, compute the predicted class probabilities.
 - Calculate the loss using the new loss function.
 - Perform backpropagation and update the model parameters.